Dhairya Chhabra

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Education

San Diego State University | *M.S in Computer Science* **Aug 2024 – May 2026**
Coursework: Data Structures and Algorithms, Machine Learning, Data Science GPA: 3.74

Savitribai Phule Pune University | *B.E in Computer Engineering* **Aug 2019 – May 2023**
Coursework: Database Management Systems, Distributed Systems, Object Oriented Programming GPA: 3.94

Technical Skills

Languages: Python, JavaScript, TypeScript, Go, Java, C++, SQL
Frameworks & Libraries: ReactJS, NextJS, NodeJS, Express, FastAPI, Spring Boot, Redux, Jest, JUnit
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Grafana
Databases: MongoDB, PostgreSQL, MySQL, Redis
System Design: Distributed Systems, Microservices, Event-Driven Architecture, Kafka, REST APIs, Scalable Architecture
AI/ML: LLMs, Transformer Architecture, RAG, Embeddings, Vector Search, OpenAI API, GenAI, Neural Networks
Soft Skills: Communication skills, Leadership, Growth Mindset, Interpersonal skills, Critical Thinking, Teamwork

Work Experience

San Diego State University - TA Software Engineer **Sept 2024 – Present**
Tech Stack: Java, Python, AWS, JUnit, FastAPI, REST APIs, SQL San Diego, CA

- **Engineered** a **RAG-based LLM pipeline** for academic Q&A by integrating **Python, FastAPI, and vector embeddings** on **AWS**, improving information retrieval accuracy by 20%.
- **Validated** performance with **JUnit** and load tests simulating concurrent users, surfacing 4+ bottlenecks pre-production.
- **Fine-tuned** and **evaluated ML models** on academic datasets using **Python and Jupyter**, benchmarking transformer architectures to improve model accuracy by 15% over baseline.

Wordcorn - Software Engineer Intern **May 2025 – Aug 2025**
Tech Stack: ReactJS, TypeScript, Python, AWS, REST APIs, GenAI, Grafana San Jose, CA

- **Engineered** end-to-end **full-stack AI learning workflows** by building **ReactJS and TypeScript** frontends backed by **Python REST APIs**, increasing user engagement by 30%.
- **Secured REST APIs** by implementing JWT authentication and role-based access control across 20+ endpoints, preventing unauthorized access to user data.
- **Developed** GenAI summarization and recommendation pipelines by integrating **LLM APIs** with scalable Python services on **AWS**, reducing content curation effort by 35%.
- **Integrated Stripe** payment gateway into platform, enabling monetization and unlocking the revenue stream for product.

AFour Technologies - Software Engineer **Mar 2023 – Mar 2024**
Tech Stack: Java, Python, ReactJS, AWS, GCP, Docker, Jenkins, CloudWatch, REST APIs Pune, India

- **Architected** backend **microservices** using **Java, Python, and REST APIs** on **AWS and GCP**, reducing API response latency by 20% and improving distributed system scalability.
- **Developed** responsive **ReactJS** dashboards consuming REST APIs from distributed backends, cutting feature delivery cycle time by 25% and improving internal tool usability for 100+ users.
- **Streamlined** releases by 50% by **building CI/CD pipelines** with **Jenkins and Docker** across microservices.
- **Diagnosed** production incidents across **distributed microservices** as measured by 35% drop in mean resolution time, by analyzing logs via **AWS CloudWatch and Grafana** dashboards.
- **Integrated Redis** caching for frequently accessed endpoints, cutting db load by 40% and improving latency by 200ms.

Digilytics Systems LLP - Software Engineer **Dec 2021 – Dec 2022**
Tech Stack: ReactJS, JavaScript, NodeJS, MongoDB, PostgreSQL, Redis, REST APIs Pune, India

- **Engineered** web platform with **ReactJS and REST APIs**, enabling **small businesses** to go digital and increase sales.
- **Optimized** data access by implementing **Redis caching** over **PostgreSQL and SQL queries** reducing repeated database calls and improving responsiveness under load by 30%.

Projects

RootLens AI | **Tech Stack**: Python, Go, Kafka, FastAPI, Grafana, Docker

- Architected a scalable observability and analysis pipeline with **FastAPI and Go microservices**.
- Integrated vector-based log retrieval and LLM-powered reasoning to **generate automated incident summaries**, and provide actionable remediation insights across distributed systems.